free flowing mind. I can think clearly and wisely when making decisions. Being clean in my room, clothing choices and maintaining a high standard encourages maturity and I'm able to carry myself as though I'm taking on leadership role. The most important thing for me is to keep develop positive visions to move forward in my life. I'm responsible for my actions and bettering myself otherwise no one else is externally able to control the bettering of my life. I hold myself accountable for each decision I make.

FC300 Portfolio 🏛

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Theme 1 - Planning & Production

Module Code	FC300
Group	Yellow
Module Title	Extended Project
Assessment Type	FC300 Portfolio CN link:
Module Tutor Name	Ms Lorreta Fung
Student ID Number	P453311
Date of Submission	
Word Count	

<![if !vml]><![endif]> *I confirm that this assignment is my/our own work. Where I have referred to academic sources, I have provided in-text citations and included the sources in the final reference list.*

<![if !vml]><![endif]> *Where I have used AI, I have cited and referenced appropriately.*

The effects of prolonged virtual reality use on postural stability and balance.

Step 1:

Create a Work Plan:

This work plan outlines the main stages and tasks of my project, showing how much time I will spend on each activity and when I aim to complete them. It provides a clear timeline to help me stay organised and ensure steady progress throughout the project.

Phase / Activity	Objectives	Tasks / Actions	Period	Estimated Duration	Expected Output / Deliverable
1. Topic Definition & Proposal Development	Define research focus, aims and rationale.	Identify focus on VR and its effects on postural stability and perception. Review guidelines. Draft and submit proposal.	Week 1–2	2 weeks	Approved project proposal with clear research claims.
2. Preliminary Research	Build a foundation of existing research.	Search academic databases (Google Scholar, JSTOR, Scopus). - Gather sources on VR, cybersickness and spatial awareness. Create annotated bibliography.	Week 3–4	2 weeks	Preliminary literature review summary (3–4 pages) and reference list.
3. Detailed Source Analysis	Identify key themes and arguments.	Critically analyse sources (e.g., Stanney et al., 2020). Identify patterns in effects such as balance disruption and distance judgement.	Week 5–7	3 weeks	Detailed literature review section.
4. Data Collection Planning	Prepare for secondary data analysis.	Select relevant studies and datasets. Decide criteria for evaluating VR effects on users.	Week 8–9	2 weeks	Research plan and methodology outline.

5. Data Collection / Evidence Gathering	Collect evidence to support research	Gather data from journals on VR effects (e.g., cybersickness, postural instability). Organise findings.	Week 10–11	2 weeks	Compile evidence and organized notes
6. Drafting Report Sections	Develop a structured research report.	Write introduction, literature review, methodology, and analysis sections. Integrate evidence and references.	Week 12	1 week	First draft of full report.
7. Review, Feedback, and Revisions	Improve the quality of the report.	Review feedback from teacher/ supervisor. Edit for clarity, structure, and academic tone. Refine arguments.	Week 13–15	3 weeks	Improved and refined report draft.
8. Final Submission & Reflection	Complete and submit the final report.	Finalise formatting, referencing, and proofreading. Submit report. Write reflection on the research process.	Week 18	Throughout	Final submitted report and reflection.
9. Reflective Presentation Preparation	Prepare to present findings.	Create presentation slides summarising research, findings, and evaluation. Practice delivery.	Week 19	1 week	Completed presentation and delivery preparation.

Step 2:

Choose a topic:

Topic area – The physiological and perceptual effects of prolonged virtual reality use.



My chosen topic focuses on the physiological and perceptual effects of virtual reality following prolonged use. It examines the potential health concerns associated with extended immersive experiences, particularly in relation to postural stability and balance judgement. Furthermore, the study explores how these effects may influence users' spatial awareness and ability to accurately interact with real world environments after VR exposure. The topic is linked to my degree in Computing and IT related concepts.

Virtual reality recently became an increasingly influential innovation in the entertainment industry particularly with games and film. Initially the idea was to examine the influence and growth of virtual reality within the wider entertainment industry, particularly in gaming and film. I chose this topic due to the rapid technological advancements in VR and its increasing visibility in mainstream media. The subject offered an opportunity to explore both technological innovation and its cultural impact. During research it became clear that focusing on the effects users experience particularly after extended periods of use would offer a more in-depth analysis on virtual reality technology and health concerns surrounding it.

1. **What was the initial idea for the project, why did you choose this topic and how did it evolve over time?**

2. My initial idea was to examine the influence and growth of virtual reality within the wider entertainment industry, particularly in gaming and film. I chose this topic due to the rapid technological advancements in VR and its increasing visibility in mainstream media. The subject offered an opportunity to explore both technological innovation and its cultural impact. During research it became clear that the gaming sector demonstrated the most significant practical integration of VR, which led to a more focused analysis within that area.

3. The questions being answered were:

1. How does prolonged VR use affect postural stability immediately after exposure?

2. Does VR use affect a user's ability to judge distances or navigate physical spaces accurately?

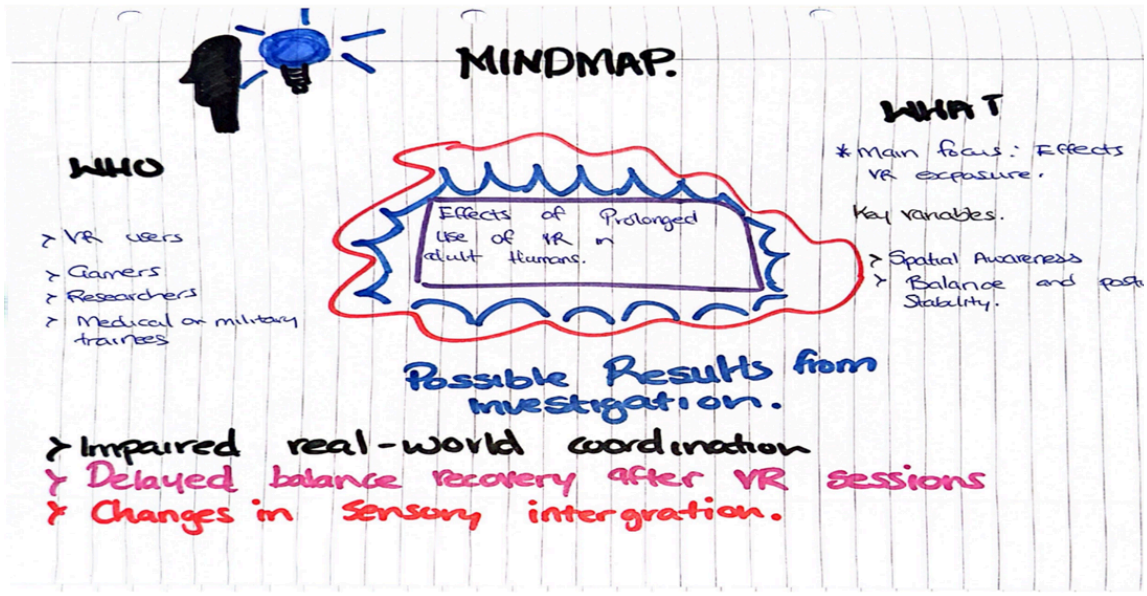
3. Do experienced VR users show fewer balance and spatial awareness disruptions than first time users?

Aim

The primary aim of this research is to investigate the physiological and perceptual effects of prolonged virtual reality use, with a particular focus on postural stability and spatial awareness. The study seeks to evaluate how extended VR exposure impacts users' balance, coordination, and ability to navigate real-world environments.

Step 3:

Mind Map:



Here is a simple Mind Map that helps me define the focus of my research topic.

It is important to truly research and understand your topic to recognize the main aspects you want to convey. Without researching, it may be difficult to be precise and maintain relevant points.

The table below highlights these four main categories, (effects, causes, challenges, and practical solutions), and the idea will build from there,

Effects	Causes	Challenges	Practical sol
Postural stability	Sensory conflict between visual input and vestibular feedback	Users may experience dizziness and reduced balance after VR use	Limit session and include r breaks
Spatial awareness/ distance judgement	Inaccurate depth perception and altered visual cues in VR environments	Difficulty judging distances or navigating real-world spaces after use	Use calibrati and gradual readaptation environments
Role of user experience	Temporary disruption of sensory integration after prolonged immersion	Reduced hand eye coordination and delayed reaction times.	Implement tra allow recovery uses.

Drafting a detailed report can feel confusing initially, especially when the focus is unclear. Things fall into place when proper research is done. Developing a firm grip on the main topic and locating solid sources to support your ideas reduces stress and confusion. Mistakes are expected however, the direction of focus can be changed, and points can be altered or scrapped along the way. It is all part of the process. It is normal and can lead to producing stronger work.

Step 4:

Formulate a specific question:

The following are my Initial and Final research questions and key terms that ensured my focus on specific points that I wanted to convey in my report:

4. **Initial research questions:** My original topic examined the growth and influence of virtual reality technology within the wider entertainment industry, particularly in film and gaming. Given the increasing popularity and commercial success of VR products, the initial aim of the research was to explore the technological advancements that have contributed to the development and widespread adoption of VR innovations. However, as the research progressed the focus was refined to investigate the physiological and perceptual effects experienced by users following prolonged exposure to VR technology. In particular, the study concentrated on two key areas: postural stability and distance judgement. Narrowing the focus in this way allowed research to produce more relevant finings and made it easier to identify appropriate academic sources.

    1. What are the key technological advancements within VR technology that have contributed to commercial success of VR products.
    2. How has the development of VR hardware and software improved the level of immersion experienced by users?
    3. What role has the gaming and film industry played in the growth and adoption of VR technology?

Final research questions:

To provide a clearer and more focused direction for the investigation, the following research questions were developed to guide the study:

4. To what extent does prolonged virtual reality (VR) use affect postural stability immediately after exposure?
5. How does VR influence a user's ability to accurately judge distances and navigate physical environments?
6. Do experienced VR users demonstrate fewer disruptions in balance and spatial awareness compared with first-time users?

Initial Research questions	Final Research questions
1. What are the key technological advancements within VR technology that have contributed to commercial success of VR products. 2. How has the development of VR hardware and software improved the level of immersion experienced by users? 3. What role has the gaming and film industry played in the growth and adoption of VR technology?	1. To what extent does prolonged virtual reality (VR) use affe stability immediately after exposure? 2. How Does VR use influence a user's ability to accurately j distances and navigate physical environments? 3. Do experienced VR users demonstrate fewer disruptions i and spatial awareness compared with first-time users? ▪

Refining my working title:

Initial title: The factors that gain virtual reality commercial success and popularity in the entertainment industry.

Final title – The effects of prolonged virtual reality use on postural stability and spatial awareness in the real world.

Step 5:

Construct a Final Plan:

Theme 2 - Information Literacy

Step 6:

Locate and Evaluate Academic Sources:

Academic sources need to be relevant, dependable, recent, and valid. To locate sources there are various online database libraries like Google scholar for example which was used during the process of locating sources. In the case of my report, academic journals were the most useful source of information.

When locating sources for my report, I applied several criteria to ensure the information was dependable and appropriate. Firstly, I considered relevance, selecting sources that directly addressed the topic question and contributed meaningful information towards my research. I also evaluated the currency of each source, prioritizing more recent publications so that the data and perspectives reflected current knowledge and developments in the field. In addition, I examined the background, institutional affiliations, and whether the material came from reputable journals, books, or established websites. Finally, I assessed reliability by comparing information across multiple sources and looking for evidence such as citations and references. Using these criteria helped ensure that the sources included in my report were credible, accurate and suitable for academic research.

Here are some key terms I used while looking up information for my topic:

Key terms (key words or phrases that will help while finding sources)

Virtual reality
Postural stability
Distance judgement
Effects of virtual reality on the vestibular system
Balance coordination after prolonged VR sessions

These key terms assisted with narrowing down research to relevant sources about effects experienced after prolonged exposure to virtual reality.

Here is an example of a source that I found particularly relevant to my report:

Step 7:

Paraphrasing Information from the sources:

Full reference of source:

Stanney, K. M., Lawson, B. D., Rokers, B., Dennison, M., Fidopiastis, C., Stoffregen, T., Weech, S. and Fulvio, J. M. (2020) Identifying causes of and solutions for cybersickness in immersive technology: Reformulation of a research and development agenda. International Journal of Human–Computer Interaction, 36(19), pp. 1783–1803. Available at: https://doi.org/10.1080/10447318.2020.1828535 (Accessed: 17 March 2026).

Step 8:

This stage of the project focused on developing initial analysis from the research gathered in previous steps. After evaluating and paraphrasing key academic sources, the next step was to begin identifying patterns, relationships, and key arguments that would form the foundation of the report.

The analysis highlighted that prolonged virtual reality (VR) use can have measurable effects on both physiological and perceptual functions. A recurring theme across multiple sources was the disruption of sensory integration, particularly the conflict between visual input and physical movement. This sensory mismatch was frequently linked to symptoms such as reduced postural stability, disorientation, and difficulty maintaining balance after VR exposure.

In addition, research findings suggested that VR can influence spatial awareness and distance judgement. Users exposed to immersive virtual environments may experience temporary changes in depth perception, which can affect their ability to accurately navigate real-world spaces. This supports the argument that VR does not only impact physical balance but also cognitive processing related to perception.

Another key point identified during the analysis was the role of user experience. Evidence suggests that individuals with prior exposure to VR may demonstrate fewer negative effects compared to first-time users, indicating that adaptation and familiarity may reduce the severity of these disruptions over time.

This stage was important in shaping the direction of the report, as it allowed the research to move beyond description and towards critical analysis. The insights gained here directly informed the structure of the main body sections and ensured that the final report would present a clear, evidence-based argument.

Step 9:

Use data to support your claims:

Step 10:

Use of appropriate paragraph structure:

PEEL (Point, Evidence, Explanation, Link) STRUCTURE:

Point: Prolonged use of virtual reality (VR) can negatively affect users' perception and balance, raising concerns about its impact on physical stability and spatial awareness.

Evidence: Stanney et al. (2020) identified that extended exposure to immersive VR environments can lead to symptoms of cybersickness, including disorientation, visual disturbances and reduced postural stability. The study highlights that these effects may persist even after the VR session has ended.

Explanation: These findings suggest that VR use can disrupt the body's ability to accurately interpret sensory information, particularly the relationship between visual input and physical movement. As a result, users may experience difficulties in maintaining balance and judging distances, which could increase the risk of errors or accidents in real-world environments.

Link: Therefore, it can be argued that while VR technology offers significant benefits, its potential impact on user perception and balance must be carefully considered, particularly when evaluating its long-term use and safety.

Step 11:

Obtain and evaluate feedback from GenAI:

Provide strong points suitable for the main body of the topic: The effects of prolonged virtual reality immersion on postural stability and spatial awareness for an

academic report.

To Deepen Your Understanding of the Topic

- "What are the key physiological effects of prolonged virtual reality use on the human body?"

- "How does extended VR exposure influence postural stability and balance in users?"
- "What factors contribute to cybersickness and spatial disorientation in virtual reality environments?"

To Get Better Sources and Evidence

- "Can you find recent academic studies on the effects of VR on balance and spatial awareness?"
- "What data is available on postural stability or coordination after VR exposure?"
- "Are there any case studies or experiments that measure the impact of VR on user perception?"

To Improve Your Report Structure and Argument

- "Can you help me build a strong and clear structure for a report on the effects of VR on perception and balance?"
- "What are the main limitations or counterarguments related to research on VR's physical effects?"
- "How can I link scientific research on VR to real-world implications for user safety and performance?"

Asking Copilot targeted questions while researching and writing can significantly improve the quality and depth of an academic report on the effects of virtual reality. These questions help to develop a clearer understanding of the topic, strengthen the use of evidence, and support a well-structured and critical analysis.

The Gen AI suggestions were helpful; they guided you in developing focused paragraphs, refining your research questions, and sharpening your thesis statement's clarity. Still, asking additional questions was important to clarify vague points, strengthen your evidence, and make sure everything aligned with your report's goals. This process of requesting and refining helped turn your initial ideas into well-supported, structured academic content.

Step 12:

Upload your sources:

Reference list:

Cambridge Dictionary (2019) Virtual reality. [online] Available at: https://dictionary.cambridge.org/dictionary/english/virtual-reality **(Accessed: 17 March 2026). Dennison, M. S. and D'Zmura, M. (2017) 'Cybersickness without the wobble: Experimental results speak against postural instability theory', Applied Ergonomics, 58, pp. 215–223. Available at:** https://doi.org/10.1016/j.apergo.2016.06.014 **(Accessed: 17 March 2026). Keshavarz, B., Hecht, H. and Lawson, B. D. (2015) 'Visually induced motion sickness: Causes, characteristics, and countermeasures', Handbook of Virtual Environments, pp. 647–697. Available at:** https://doi.org/10.1201/b17360 **(Accessed: 17 March 2026). Lawson, B. D. (2014) 'Motion sickness symptomatology and origins', Handbook of Virtual Environments, pp. 531–599. Available at:** https://doi.org/10.1201/b17360 **(Accessed: 17 March 2026).**

Stanney, K. M., Lawson, B. D., Rokers, B., Dennison, M., Fidopiastis, C., Stoffregen, T., Weech, S. and Fulvio, J. M. (2020) Identifying causes of and solutions for cybersickness in immersive technology: Reformulation of a research and development agenda. International Journal of Human–Computer Interaction, 36(19), pp. 1783–1803. Available at: https://doi.org/10.1080/10447318.2020.1828535 **(Accessed: 17 March 2026). Weech, S., Kenny, S. and Barnett-Cowan, M. (2019) 'Presence and cybersickness in virtual reality are negatively related: A review', Frontiers in Psychology, 10, p. 158. Available at:** https://doi.org/10.3389/fpsyg.2019.00158 **(Accessed: 17 March 2026).**

Theme 3 - Reflection & Development

Reflection

This section reflects my personal experience creating this portfolio.

Challenges

- I initially struggled to find a topic that is both interesting and academically appropriate.
- I had difficulties understanding complex academic language in journal articles.
- Finding a balance between a topic that is too broad or one that is too narrow.
- Difficulty identifying enough reliable academic sources to support the topic.
- Difficulty using appropriate terminology to present my information in a professional manner.
- Uncertainty about how to turn a general idea into clear research questions.
- Lack of confidence in whether the chosen topic would meet assessment criteria.

- Completing a portfolio is a new experience. I need to document evidence and show my progress. Continuous effort is required, and I need to maintain discipline to complete my work.

Successes

- Successfully narrowing down the topic to a clear and focused area of research.
- Improved ability to find and use credible academic sources.
- Gained confidence in analysing and interpreting research findings.
- Developed a better understanding of academic writing structure.

- Improved referencing skills and awareness of academic integrity.
- Became more independent in research rather than relying heavily on AI.
- Better time management and organisation throughout the project.
- Increased understanding of the chosen topic (VR effects on posture and balance).
- Built confidence in completing a full academic report from start to finish.

Education

Nottingham Trent International College

Sep 01, 2025 - Present

Great Lecturers, inclusive and deeply engaging experience. Currently doing Computing (with foundation year) BSc (Hons)

Chengelo School

Aug 01, 2016 - Nov 01, 2022

Great teachers, amazing learning environment. I completed the IGCSE syllabus with pass grades across all subjects. Attended ISAZ sports tournaments representing the school team in football and hockey.

Learn more about me by visitng my ePortfolio!
theCN.com/LB1370